

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: SUB-STRAND 2.1 SIMILARITY AND ENLARGEMENT

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Think back to the Kisumu school plan labelled Scale 1 : 200 and the two water tanks, one of which holds 8 times as much water as the other. Answer every section in your own words. Use the evidence you collected in Lessons 1 to 8, including your measurements, tables and models. Show all your working for calculations and write full sentences when asked to explain. Use the words linear scale factor (L.S.F), area scale factor (A.S.F), volume scale factor (V.S.F) and centre of enlargement correctly. Your teacher will mark your work against the rubric.

Section 1: What makes figures similar, and what does Scale 1 : 200 mean?

The Kisumu school plan is labelled Scale 1 : 200. A classroom block that is 10 m wide appears on the plan with a certain width. (a) State the two conditions that must be true for two figures to be similar. (b) Calculate the width of the classroom block on the plan in centimetres, and write the L.S.F as a fraction. (c) Explain why every window, door and corridor on the plan is still in the right proportion even though the whole school compound fits on one A3 sheet.

(a) Two figures are similar if (1) all corresponding angles are equal and (2) all corresponding sides are in the same ratio, which is the Linear Scale Factor. (b) $10 \text{ m} = 1000 \text{ cm}$. Plan width = $1000 \div 200 = 5 \text{ cm}$. L.S.F = image length $\div$ object length = $5 \div 1000 = 1/200$. (c) The same L.S.F of $1/200$ is applied to every length in the compound, so all corresponding sides are in the same ratio. The angles stay equal, so the shape of every part is preserved. The plan is therefore a similar figure of the real compound: it is much smaller, but every part is in the right proportion.

Section 2: Centre of enlargement and constructing images

A triangle ABC represents a classroom block. It is enlarged from a centre O with L.S.F $k = 2$ and then separately with $k = -2$. (a) Explain how you can find the centre of enlargement when

(a) Draw a ray line through each pair of corresponding vertices (A and A', B and B', C and C'). All the lines meet at one point, and that point is the centre of enlargement O. (b) For $k = 2$ the image lies on the same side of O as the object, on the same ray, at twice the distance from O. Every image side is 2 times the object side. For $k = -2$ the image lies on the opposite side of O, still at twice the distance (lengths are scaled by $|k| = 2$), and it is turned through 180° about O. (c) With the origin as the centre, Image = $k \times$ Object. For $k = 2$: image of A = $2 \times (2, 1) = (4, 2)$. For $k = -2$: image of A = $-2 \times (2, 1) = (-4, -2)$, which is on the opposite side of the origin, in the third quadrant.

you are given an object and its image. (b) Describe where the image is placed, and how it is oriented, for $k = 2$ and for $k = -2$. (c) The point A is at (2, 1) and the centre is the origin. Find the image of A for $k = 2$ and for $k = -2$ using the formula Image = Centre + $k \times$ (Object – Centre).

Section 3: How area changes under enlargement

In Lesson 5 you compared similar rectangles and triangles. (a) A student predicts that doubling every length of a rectangle doubles its area. Use a rectangle 3 cm by 2 cm and its enlargement with L.S.F 2 to test this prediction. (b) State the rule connecting A.S.F and L.S.F and explain why it works. (c) The base of a 2 kg Unga Pembe flour packet has an area 4 times that of the 1 kg packet. What is the L.S.F between the two bases? (d) At Scale 1 : 200, how many times smaller is an area on the plan than the real area?

(a) Original rectangle: $3 \times 2 = 6 \text{ cm}^2$. Enlarged rectangle: 6 cm by 4 cm, so area = 24 cm^2 . The area ratio is $24 \div 6 = 4$, not 2. The prediction is wrong: area quadruples. (b) $\text{A.S.F} = (\text{L.S.F})^2$. Every length is multiplied by k , and area is found by multiplying two lengths, so the area is multiplied by $k \times k = k^2$. (c) $\text{A.S.F} = 4 = (\text{L.S.F})^2$, so $\text{L.S.F} = 2$. (d) $\text{L.S.F} = 1/200$, so $\text{A.S.F} = (1/200)^2 = 1/40\,000$. Areas on the plan are 40 000 times smaller than the real areas.

Section 4: Solving the water tank puzzle

The school has two water tanks of identical shape. The larger tank is twice as tall as the smaller tank. (a) State the rule connecting V.S.F and L.S.F. (b) Use the rule to work out how many times as much water the larger tank holds as the smaller tank, and explain why your answer is not 2. (c) Copy and complete this table for pairs of

(a) $\text{V.S.F} = (\text{L.S.F})^3$. (b) The tanks are similar and the larger is twice as tall, so $\text{L.S.F} = 2$. Then $\text{V.S.F} = 2^3 = 8$, so it holds 8 times as much water, not 2 times. Length, width and height are all doubled, so volume is multiplied by $2 \times 2 \times 2$. (c)

L.S.F	A.S.F	V.S.F
2	4	8
3	9	27

similar solids, then show the calculations for the last row.

L.S.F	A.S.F	V.S.F
2		
3		

(d) A sufuria has a capacity of 2 litres. A similar sufuria has an L.S.F of 3 compared with it. What is the capacity of the larger sufuria?

For L.S.F = 3: A.S.F = $3^2 = 9$ and V.S.F = $3^3 = 27$. (d) V.S.F = $3^3 = 27$, so capacity = $2 \times 27 = 54$ litres.

Section 5: Answering the driving question

Driving question: When you scale up a shape or solid, such as doubling the size of a water tank or enlarging a building plan, what exactly happens to its lengths, its area and its volume, and how can we use this to solve real problems in our community? (a) Write a paragraph that answers the driving question using the terms L.S.F, A.S.F and V.S.F and referring to the school plan and the water tanks. (b) Give one new real-life Kenyan example, different from the ones used in class, where you would use these scale factors, and show a calculation. (c) Complete this reflection: 'In Lesson 1 I used to think ... Now I know ...'

(a) When a shape or solid is enlarged by a linear scale factor k , every length is multiplied by k , every area is multiplied by k^2 and every volume is multiplied by k^3 . The Kisumu school plan at Scale 1 : 200 has L.S.F = $1/200$, so each length is 200 times smaller, which keeps the proportions of windows, doors and corridors correct, while areas are 40 000 times smaller. The larger water tank is twice as tall as the smaller one, so L.S.F = 2 and V.S.F = $2^3 = 8$, which is why it holds 8 times as much water. We can use these rules to calculate map distances, plan areas, tank capacities and packaging sizes. (b) Example: A farmer in Kericho has a grain store that holds 500 kg of maize. A similar store is built with all lengths 2 times as large. V.S.F = $2^3 = 8$, so it holds $8 \times 500 = 4000$ kg. The roof area needed is $2^2 = 4$ times as much. (c) In Lesson 1 I used to think that doubling the size of a shape doubles its area and volume. Now I know that doubling the lengths multiplies the area by 4 and the volume by 8, because area scales by k^2 and volume by k^3 .

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of similarity and scale	States both conditions for similar figures and correctly calculates the plan width and L.S.F of 1/200, explaining clearly why proportions are preserved.	States both conditions and calculates the L.S.F correctly, with a partly complete explanation of proportion.	States only one condition or makes errors in converting units or finding the L.S.F.
Enlargement and centre of enlargement	Explains how to find the centre using ray lines, describes positive and negative k accurately, and calculates image coordinates correctly.	Describes the centre and the effect of k with minor gaps, and gets most coordinates correct.	Cannot describe how to find the centre or confuses positive and negative k.
Area and volume scale factors	Correctly applies $A.S.F = (L.S.F)^2$ and $V.S.F = (L.S.F)^3$ in all calculations and explains why each rule works.	Applies both rules correctly in most calculations with a limited explanation.	Treats area or volume as scaling by k, or makes frequent calculation errors.
Explanation and use of evidence	Connects the school plan and water tanks to the scale factor rules with clear, complete reasoning and accurate use of terms.	Connects the phenomenon to the rules with generally accurate terms but some reasoning left implicit.	Gives statements with little reasoning or little link to the phenomenon.
Real-life application and reflection	Gives a new, realistic Kenyan example with a correct calculation and a clear 'used to think / now I know' reflection.	Gives a suitable example with a mostly correct calculation and a brief reflection.	Example is unsuitable or the calculation is incorrect, and the reflection is missing or vague.

ATTRIBUTION AND LICENSING

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